

ARTIFICIAL INTELLIGENCE II

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Classification using Prompting Techniques

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Prompting: A Reminder

Prompting is the practice of communicating with a large language model through natural language instructions.

No Weight Updates

The model's parameters remain frozen — only the input changes.

Leverages Existing Knowledge

Guides behavior through structure and content of the input alone.

Natural Language Interface

Instructions are written in plain language, not code or configuration.



Fine-Tuning vs Prompting

Dimension	Prompting	Fine-Tuning
Model weights	Unchanged	Updated
Data required	Few examples (or none)	Large labeled dataset
Cost	Low — inference only	High — training compute
Flexibility	Switch tasks instantly	One model per task (typically)
Performance ceiling	Limited by base model knowledge	Can specialize deeply
Failure mode	Model ignores instructions	Overfitting, catastrophic forgetting

 The core trade-off: prompting is fast and cheap but bounded by the base model's knowledge, while fine-tuning can push performance further but requires data, compute, and a separate model per task.

Models used for prompting

Encoder-Only Models

Models like **BERT** and **RoBERTa** read the entire input bidirectionally and produce a rich contextual representation. They are trained to understand text — filling in masked tokens, classifying sequences, comparing sentences. They do not generate new text; they output a vector.

Decoder-Only Models

Models like **GPT**, **Qwen**, and **LLaMA** are trained to predict the next token. Generation is their native operation — given a prompt, they continue it. Prompting works by placing instructions in the input and letting the model continue from there.

The Qwen 3.5 Family

Released in February–March 2026 by Alibaba Cloud, **Qwen 3.5** is a family of open-weight decoder-only models — exactly the kind we prompt. The small series (0.8B, 2B, 4B) is the focus of this tutorial.

- ✅ These variants are small enough to run on standard academic hardware, yet deliver a exceptional performance — making them ideal for a systematic comparison of prompting strategies across model sizes.

Prompt Structure

A prompt is the input we pass to a language model. It can contain up to four components. Not all four are required in every prompt — a minimal prompt may contain only an instruction. The more structured the task, the more components tend to be useful.



Instruction

What you want the model to do — the core directive that shapes the entire response.



Input Data

The specific instance to process — e.g., the question–answer pair in a classification task.



Context

Background information the model should take into account when generating its response.



Output Indicator

A signal that tells the model what format or type of response is expected.

System Prompt vs User Prompt

Instruction-tuned models typically distinguish between two roles. Keeping these two roles separate is good practice: the system prompt stays fixed across all examples, while the user prompt changes with each instance.

System Prompt

Sets the model's behavior, persona, or constraints *before* the conversation begins. This is where you define the task, the output format, and any rules the model should follow throughout.

User Prompt

The per-instance input. In a classification task, this is where the specific question–answer pair goes. It changes with each new example while the system prompt remains fixed.

Output Parsing

In the early days of prompting, extracting a structured answer from a model's response was a significant engineering challenge.

Models would often:

→ **Ignore Formatting Instructions**

Responding in free-form prose regardless of the requested structure.

→ **Bury the Answer**

Producing the correct answer inside an explanation: *"I think the answer is C, because..."*

→ **Inconsistent Formatting**

Varying output format across different runs or examples of the same task.

→ **Hallucinate Labels**

Inventing additional labels or categories not present in the original instruction.

- ✅ Today, well instruction-tuned models — even small ones like Qwen3.5-0.8B — have largely internalized the ability to follow formatting instructions reliably, reducing the need for complex output handling pipelines.

Parameters

Parameter selection should be guided by three factors: the model you are using, the task at hand, and the prompting method you have chosen.

What output length am I expecting?

Short classification labels require very different max-token settings than long-form generation. Calibrate your length budget to the task.

Do I want my model to be creative or precise?

Temperature and top-p control the randomness of sampling. Lower values yield more deterministic, focused outputs.

Read the model card for suggested values and models' particularities.

Studying Advice & Good Academic Practices

On Reading Papers

If you are implementing a prompting method you are not fully familiar with, a practical approach is to upload the original paper to a tool like **NotebookLM**, watch the auto-generated overview to build intuition quickly, and then return to the paper to extract the implementation details you need. This is a reasonable trade-off between time invested and depth of understanding achieved.

On AI Assistance Disclosure

A mark of rigorous academic practice is transparency about the tools used in your work. You are encouraged to include a short **AI Assistance paragraph** at the end of your report describing which models you used and what you used them for — for example: brainstorming, coding, result analysis, or report writing.

- ❏ This is not mandatory, but recommended. Being explicit about AI assistance does not diminish your work; it reflects honest and thoughtful scholarship.

Q&A

Thank You!

Questions, comments, and discussion are welcome.